

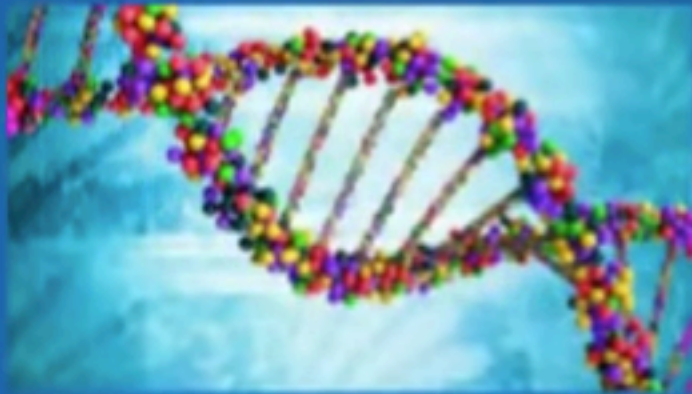
Giới Thiệu Neural Networks

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Knowledge source?

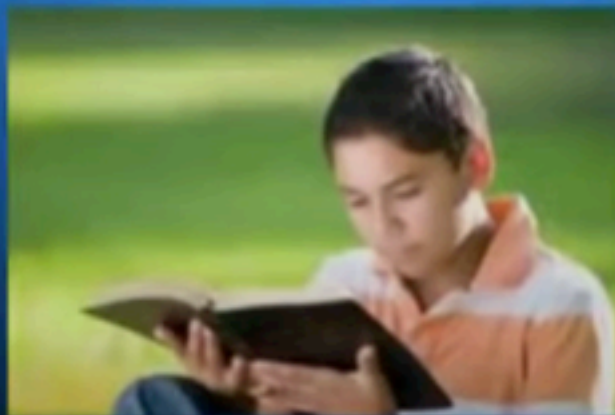
Evolution Tiến hoá



Experience

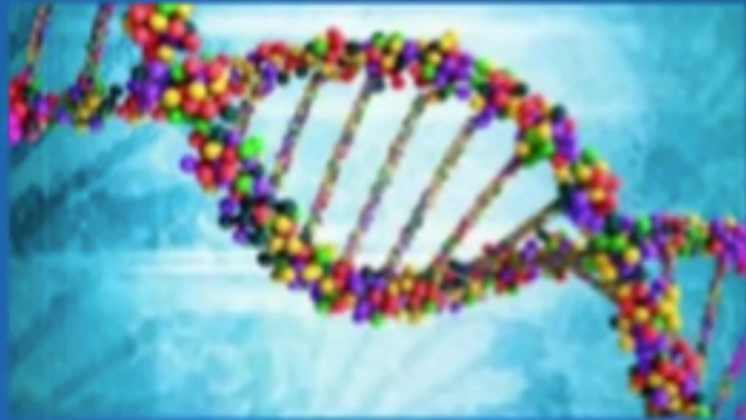


Culture



Why computing?

Evolution



Experience



Culture



Computers



Computer vs. Human

- **Computer:**

- ✓ Computing: billion of calculations / s
- ✓ Remember: 100s GB
- ✓ Do what you tell it to do -> Programming -> instructions

- **Human:**

- ✓ Observe nature
- ✓ Imagine
- ✓ Learn
- ✓ Calculation?
- ✓ Remember important features?
- ✓ Interact with environment and action independently

Examples

	label	message
0	ham	Go until jurong point, crazy.. Available only ...
1	ham	Ok lar... Joking wif u oni...
2	spam	Free entry in 2 a wkly comp to win FA Cup fina...
3	ham	U dun say so early hor... U c already then say...
4	ham	Nah I don't think he goes to usf, he lives aro...

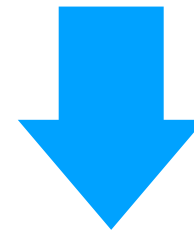
Intelligent Machine

(1) Give machine set of rules:
programming

(2) Learning from data
(**Machine learning**)

Máy tính tự động khái
hoá các tri thức ẩn từ
dữ liệu -> thực hiện
một phép phán đoán và
hành động phù hợp

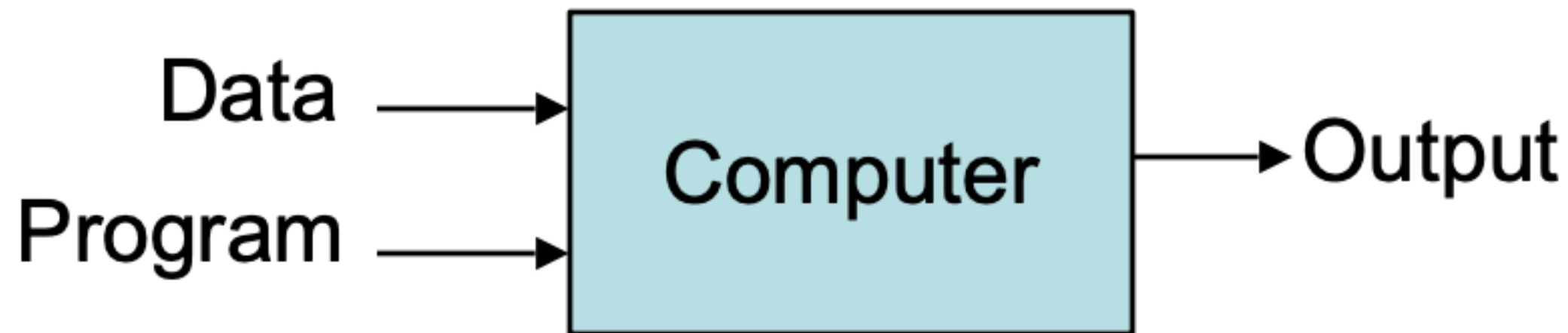
INPUT
Information
(X)



OUTPUT
Prediction (y)
Action (0, 1)

Data processing

Traditional Programming



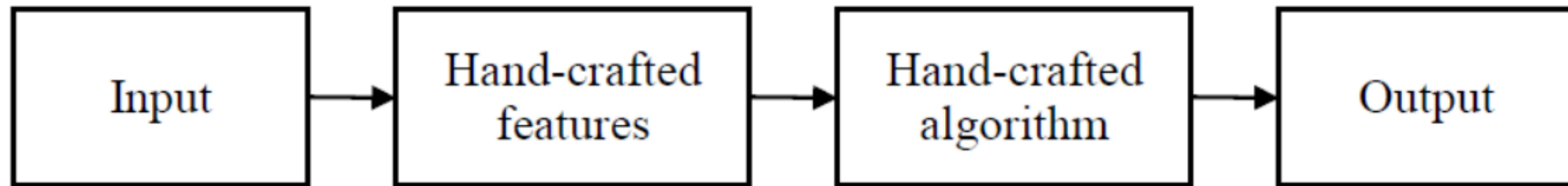
Machine Learning



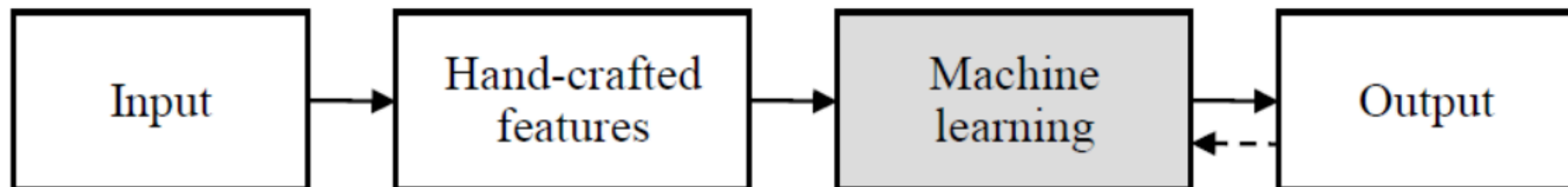
Dạy máy tính cách học

Làm thế nào để computer học?

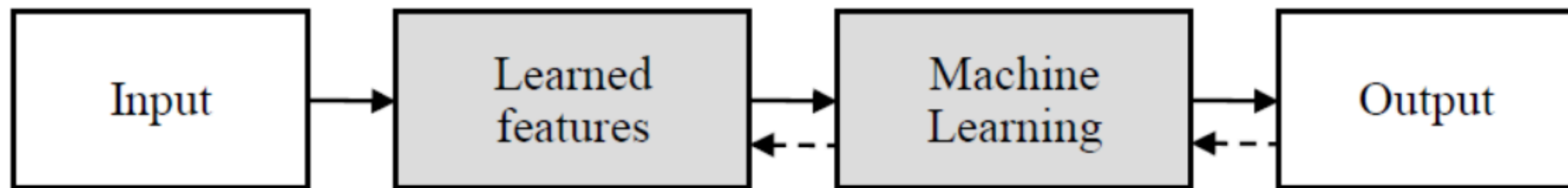
Traditional ML vs DL



(a) Traditional

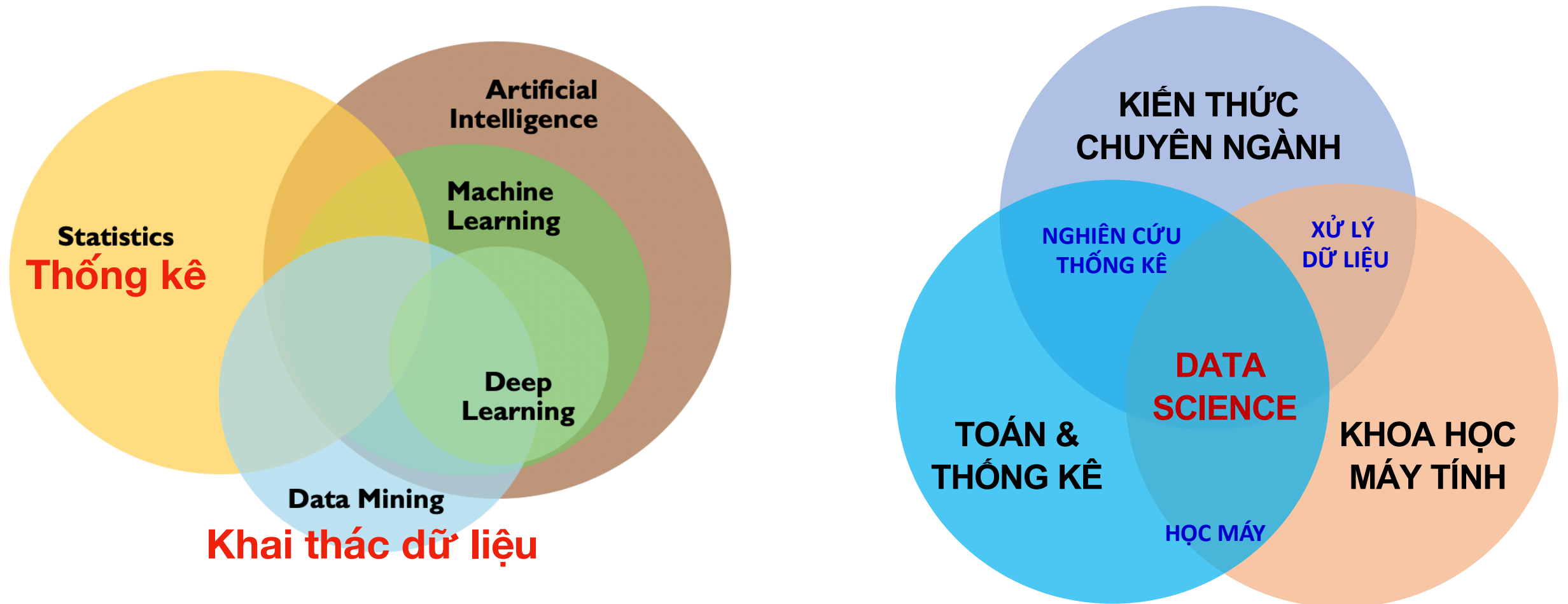


(b) Classic machine learning pipeline

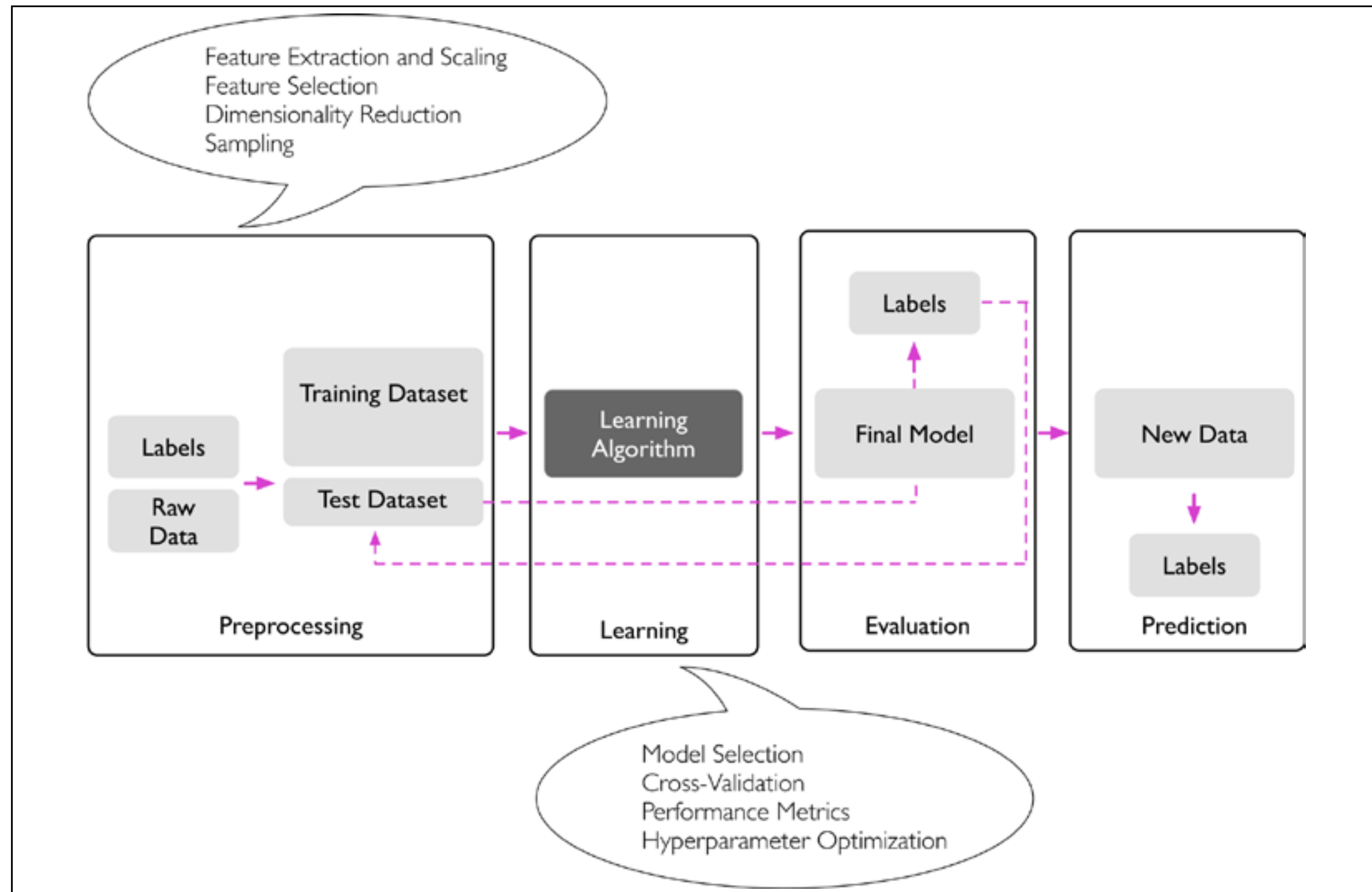


(c) Deep learning pipeline

Deep Learning và Trí tuệ Nhân tạo



Roadmap for Building Machine Learning Systems



Data-Driven Approach

(1) Problem setting

(2) Data collection

$$D = \{(x_i, y_i), i = 1, 2, \dots, m\}$$

(3) Modeling and Training Models

```
def train(images, labels):  
    # Machine learning!  
    return model
```

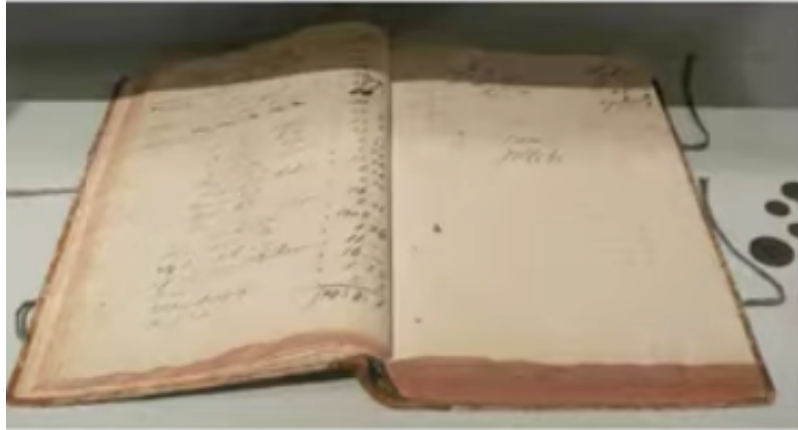
(4) Model selection

(5) Deploy suitable model (Using the best model to make prediction)

```
def predict(model, test_images):  
    # Use model to predict labels  
    return test_labels
```

Data, Information, and Knowledge

Data – raw facts
about the world



Information – Processed
or summarized raw facts
that can be used in
decision-making

Knowledge –
Information that
leads to valuable
actions

Data Driven World

3 Important Statistics About How Much Data Is Created Every Day



1 How much data is generated every minute?

Source: Domo

 **41,666,667**

messages shared
by WhatsApp users

 **1,388,889**

video / voice calls made
by people worldwide

 **404,444**

hours of video streamed
by Netflix users

 **347,222**

stories posted by Instagram users

 **150,000**

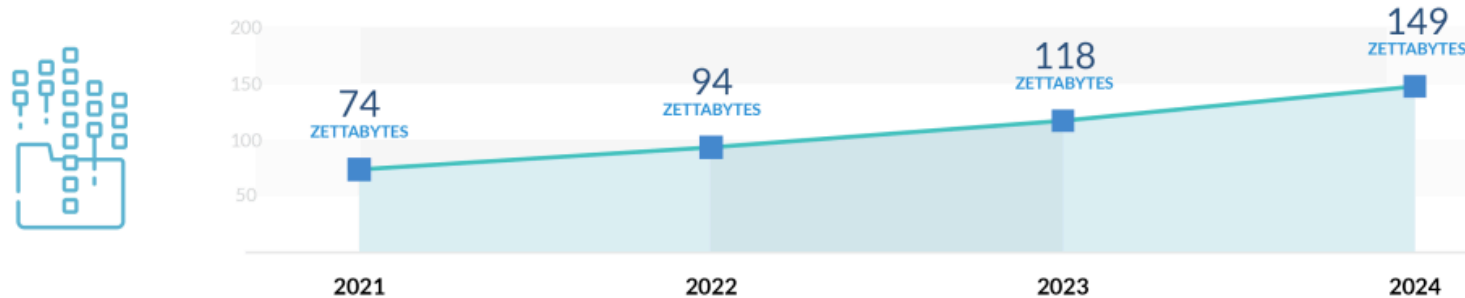
messages shared by Facebook users

 **147,000**

photos shared by Facebook users

2 Estimated Data Consumption from 2021 to 2024

Source: IDC / Statista



3 Data Growth in 2021

Sources: TechJury, Internet Live Stats, Cisco, PurpleSec

 **2 TRILLION**

searches on Google by the end of 2021

 **1.134 TRILLION MB**

volume of data created every day

 **3,026,626**

emails sent every second, 67% of which are spam

 **278,108 PETABYTES**

global IP data per month by the end of 2021

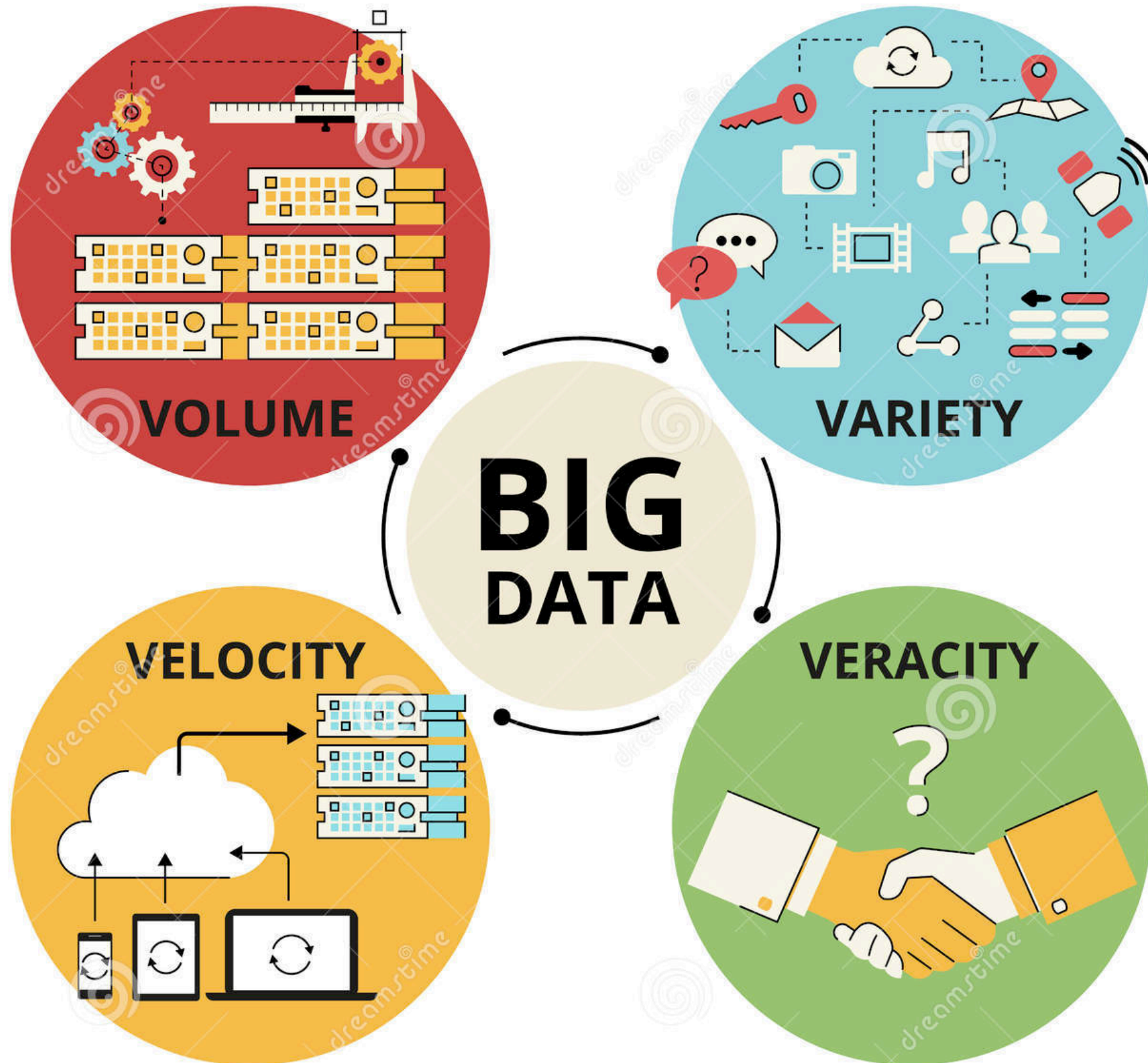
 **230,000**

new malware versions created every day

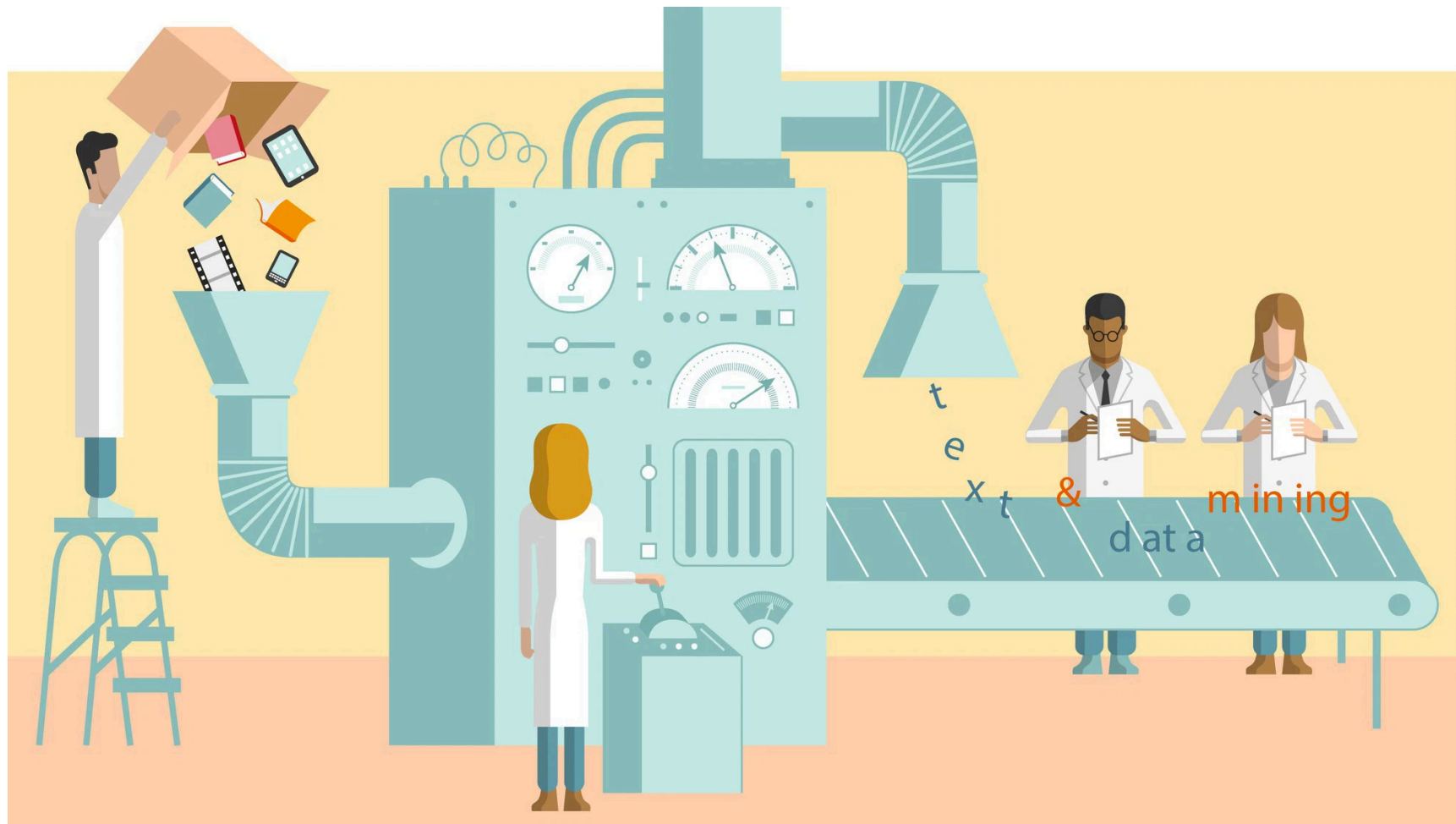
 **82%**

share of video in total global internet
traffic at the end of 2021

Big Data: 4V



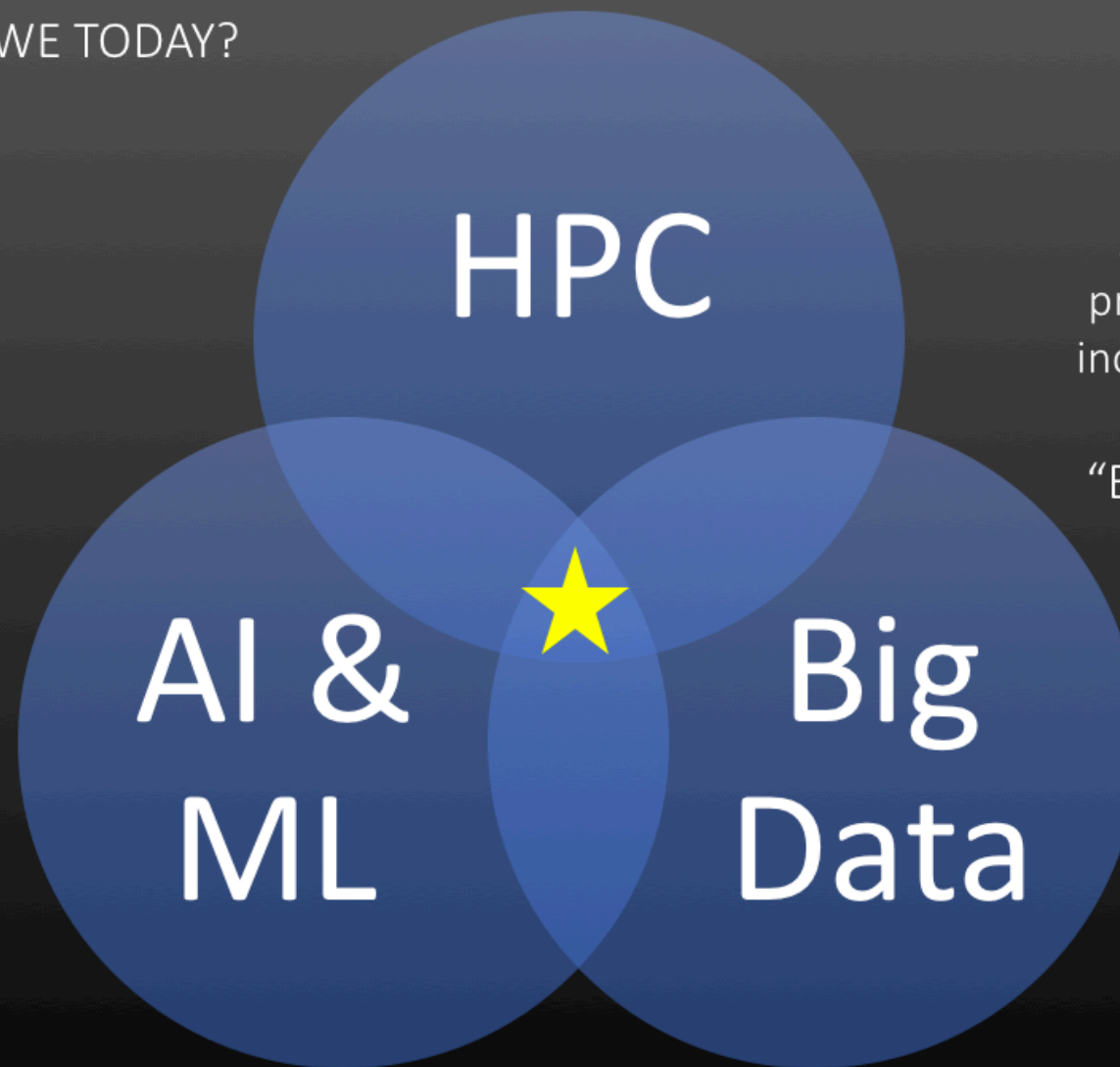
Data mining



Data mining = extraction of actionable information from (usually) very large datasets, is the subject of extreme hype, fear, and interest.

HPC - AI&ML - BigData

WHERE ARE WE TODAY?



Sweet spot that
precipitates the 4th
industrial revolution

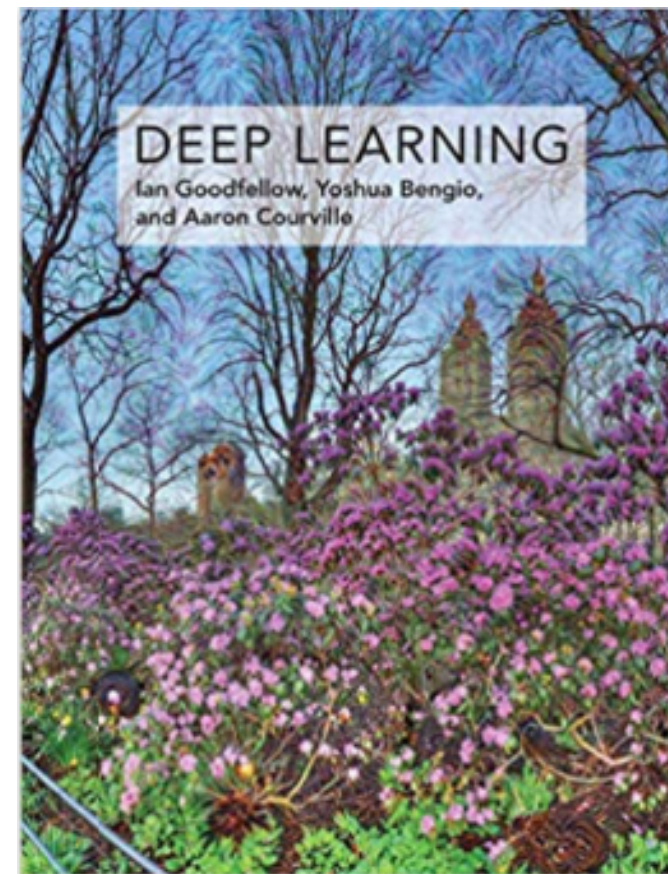
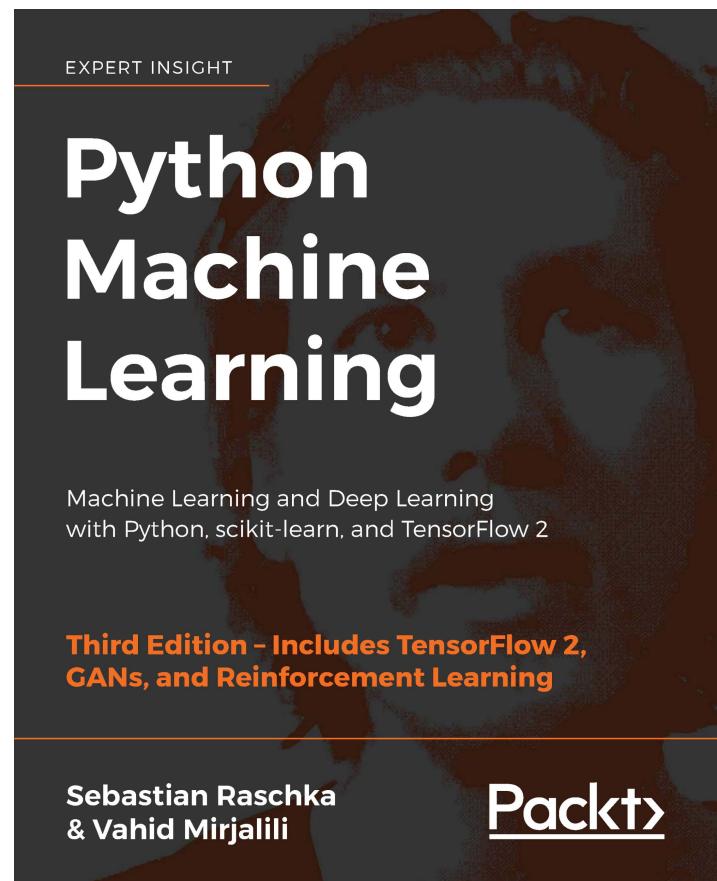
“Big Technology”

Where technologies meet new dimensions of innovation open up, which previously belonged to the realm of science fiction.

What You are Learning

- Neural Networks
- Convolutional Networks
- Recurrent Neural Networks
- Build Deep Models with Tensorflow
- Build Deep Models with Pytorch

Where to Learn



<https://www.deeplearningbook.org>

Đánh giá học phần

- Điểm chuyên cần: 10%
- Điểm bài tập: 30 %
- Điểm thi: 60%

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